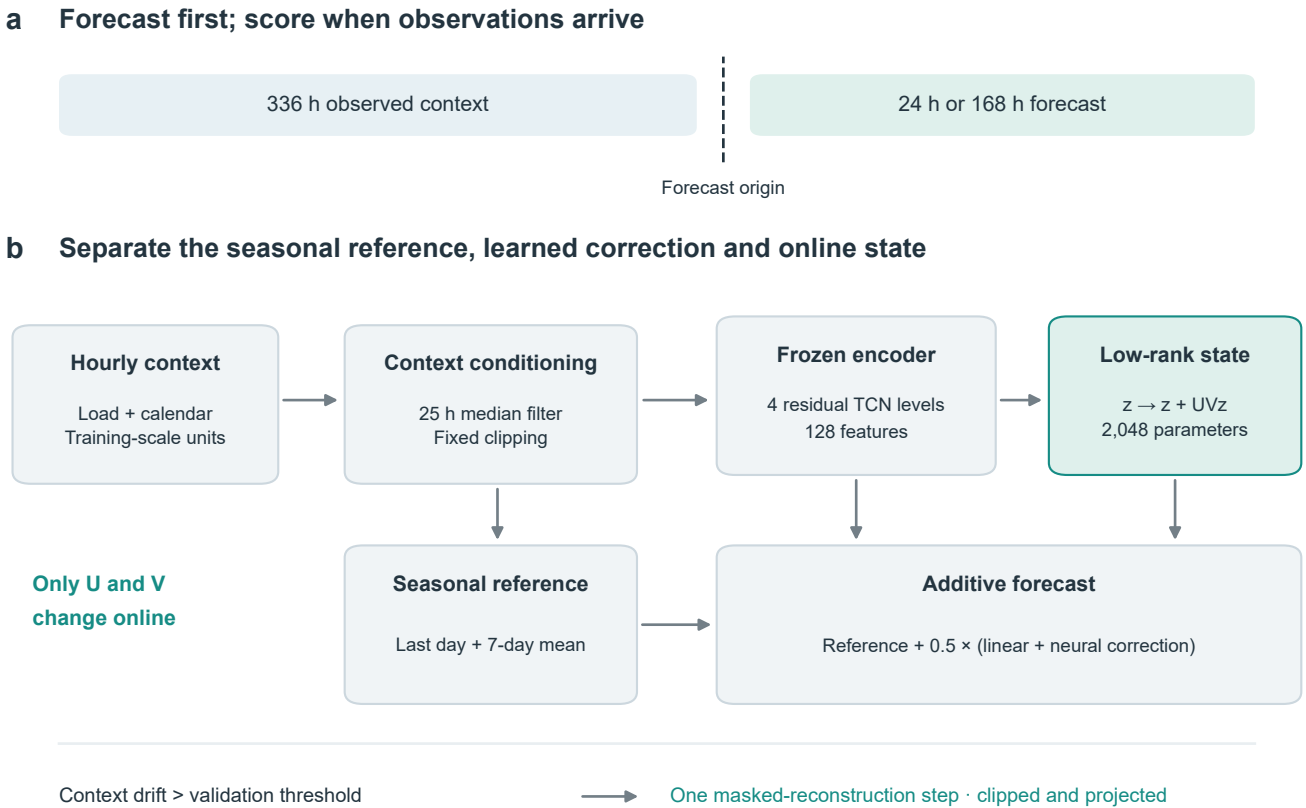
a
Forecast first; score when observations arrive
336 h observed context
24 h or 168 h forecast
Forecast origin
b
Separate the seasonal reference, learned correction and online state
Hourly context
Load + calendar
Training-scale units
Context conditioning
25 h median filter
Fixed clipping
Frozen encoder
4 residual TCN levels
128 features
Low-rank state
z → z + UVz
2,048 parameters
Only U and V change online
Seasonal reference
Last day + 7-day mean
Additive forecast
Reference + 0.5 × (linear + neural correction)
Context drift > validation threshold
One masked-reconstruction step · clipped and projected